



**Verified Carbon
Standard**

IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Tasman Environmental Markets

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Prepared By	Tasman Environmental Markets Asia Pacific
Contact	<i>Dr. Gabriel Machovsky-Capuska, Head of Science and Sustainability TEM</i> <i>15th Floor</i> <i>31 Market Street,</i> <i>Sydney, NSW 2000, Australia</i> <i>Email: gabriel@tem.com.au</i>

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Improved Cookstove Grouped Project in Papua New Guinea by Tasman Environmental Markets Asia Pacific (“TEM”) is a grouped project activity that involves the distribution of fuel efficient improved cookstoves (ICS) within the boundary of Papua New Guinea (PNG). The ICS disseminated through this project will replace open-fired wood-fueled cooking methods.

The project activities included in this grouped project activity will occur in geographically defined regions within the project boundary of PNG, starting first within the Southern Highlands Province (SHP), Western Highlands Province (WHP), Jiwaka Province (JP) and Western, Eastern Highlands Province (EHP).

For more information on exact coordinates, refer to Section 1.12 of this document.

A summary description of the technologies/measures to be implemented by the project:

The technology of the project activity involves the use of Improved Cookstoves (ICS), whereby it is designed as a stainless-steel insulated combustion chamber, with a high thermal combustion efficiency (over 25%). This highly effective stove uses less biomass fuel and emits less smoke than open-fired wood-fueled cooking methods. The implementation of the ICS is estimated to reduce the amount of wood cutting, as well as the amount of time spent by end-users foraging for firewood.

In the baseline scenario prior to the implementation of the project, there would be continued wood cutting practices and use of non-renewable wood fuel by the target population to meet the thermal energy needs for cooking. With ICS technology, woody biomass would be more efficiently combusted, hence emitting lower carbon emission and in turn generating GHG emission reductions.

Presently, a total of 40,000 ICS is planned to be distributed within Phase I of the project and will eventually scale up to 100,000 ICS within the targeted Highlands Provinces in later phases. The ICS number will vary during actual implementation and same will update during registration and verification of project activity. The average annual and total GHG emission reduction from the project is expected to be **572,936 tCO₂e** and a total of **5,729,361 tCO₂e** for the group project over the entire crediting period.

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Verification	04-04-2023 - Current	Verified Carbon Standard	Earthood Services Private Limited	-

Total				
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1.2 Sectoral Scope and Project Type

The project is categorized as following:

- Sectoral Scope 03 – Energy demand
- Type II – Energy efficiency improvement projects

1.3 Project Eligibility

The grouped project activity involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program.

1.4 Project Design

The project is a grouped project activity (GPA).

This document will also be known as the GPA Design Document (GPA-DD).

Eligibility Criteria

For the inclusion of new project activity instances, the project proponent shall ensure that each Project Instance (PI) included:

S/N	Criterion	Applicability	Evidence
1	Meets the applicability conditions set out in the methodology applied to the project.	New PIs will meet the applicability conditions set out in Section 4 of the methodology	- GPA-DD
2	Uses the technologies or measures specified in the project description.	Only ICS that conforms with the grouped project description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	- ICS manufacturer's technology description.

3	Applies the technologies or measures in the same manner as specified in the project description.	ICS distributed in the Pls will adhere to the grouped project description.	- ICS manufacturer's Technical Specifications - Invoices of Purchase
4	Are subjected to the baseline scenario determined in the project description for the specified project activity and geographic area.	New Pls will be installed only in regions within the geographic borders of PNG subject to the same baseline scenario determined in Section 3.4 in this Project Description.	- Global Position System (GPS) coordinates captured from each end user will demonstrate the location of the project. - Virtual geological log of cookstoves in monitoring database.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	All new Pls will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability	- Analysis on PNG government programs or policies for cookstoves conducted in Section 3.5. - The End User Agreement will confirm that the TEM "installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits".

		conditions of the methodology.	
6	Occur within one of the designated geographic areas specified in the project description.	New PIs will only occur within the boundary of PNG.	- GPS Coordinates of ICS end user.
7	Comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	New PIs will comply with at least one complete set of eligibility criteria for the inclusion into the GPA.	- GPA-DD
8	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	New PIs will demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body through the inclusion of sufficient technical, financial, geographic, and other relevant information within the monitoring report.	- Sampling Plan - Monitoring Report
9	Be validated at the time of verification against the applicable set of eligibility criteria.	New PIs will be validated at the time of verification against the applicable set of eligibility criteria.	- Monitoring Report
10	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	New PIs will provide evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance.	- The End User Agreement

11	Have a start date that is the same as or later than the grouped project start date.	The start date of new PIs will be after 01/12/2023, the start date of the GPA.	- ICS Purchase Invoices
12	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4.4) and new instances are eligible for crediting from the start of the next verification period.	New PIs will be eligible for crediting from the start date of the instance through to the end of the project crediting period (only).	- Monitoring Report

1.5 Project Proponent

Organization name	Tasman Environmental Markets
Contact person	Dr. Gabriel Machovsky Capuska
Title	Head of Science and Sustainability
Address	15th Floor, 31 Market Street, Sydney, NSW 2000, Australia
Telephone	+61 3 5977 1286
Email	gabriel@tem.com.au

1.6 Other Entities Involved in the Project

TEM Asia-Pacific Pte Ltd is the sole entity involved in the project.

1.7 Ownership

The project ownership is with TEM Asia-Pacific Pte Ltd.

TEM has also consulted with national authorities such as the PNG Forest Authority, wherein a Letter of Non-Objection to the project activity has been received.

During the registration of ICS, the participating households will sign a declaration to transfer the ownership rights of the carbon assets generated from this project to the project proponent.

1.8 Project Start Date

01/12/2023 (The estimated date of commissioning of ICS Phase I).

1.9 Project Crediting Period

Crediting Period: Fixed

Total Number of Years: 10 Years

Start and End Date: 01/12/2023 to 31/11/2032.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	X

The emission reduction table below accounts for the scaling up of the project over the 10-year crediting period, starting from 40,000 ICS in the first year, up to 100,000 ICS in the third year:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	308,758.98
Year 2	600,488.88
Year 3	722,497.00
Year 4	684,511.55
Year 5	648,500.54
Year 6	614,362.23
Year 7	582,000.05
Year 8	551,322.41

Year 9	522,242.39
Year 10	494,677.55
Total estimated ERs	5,729,361.59
Total number of crediting years	10
Average annual ERs	572,936.16

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient ICS to replace the baseline cookstoves in households. Prior to the implementation of the project activity, households use non-renewable woody biomass for their fuel needs. The baseline appliance being replaced by the project is an open fire (typically based on the three-stone concept).

List of facilities, systems, and equipment in operation under the existing scenario prior to the implementation of the project:

In the baseline scenario, no improvement projects relating to efficiently combusting woody biomass for cookstoves had been implemented, as such, wood fuel use has steadily been increasing as economic pressure has been increasing. The distance to wood harvesting sites increases ever more as fuel demand increases.

The ICSs reduce GHG emissions from biomass burning through improved combustion efficiency of wood fuel and decreased wood fuel consumption. This is achievable through improved heat transfer to cooking pots through the insulated walls and design of the stove top as compared to the baseline scenario and traditional cooking methods over open- and three-stone fires, wherein woody biomass would be burnt at a lower efficiency, incurring higher GHG emissions.

List and arrangement of main manufacturing technologies and equipment involved:

The project envisages to use the highest performance ICS in the market known as Kuniokoa stove with an adjustable pot skirt (see figure 1 and table 1).



Figure 1. Kuniokoa cookstove with and without adjustable pot skirt.

Table 1. Technical description of Kuniokoa cookstove based on manufactured specifications.

Parameter	Description
Thermal efficiency with pot skirt	51.3%
Efficiency	Tier 4
Emissions	Tier 3-4
Dimensions	32.1 cm height, 28.2 cm diameter
Construction materials	▪ Stove Body: Galvalume ▪ Pot Rest: Stainless Steel ▪ Pot Skirt: Stainless Steel ▪ Burning Chamber: Stainless Steel ▪ Fuel Feeding Door: Stainless Steel ▪ Stick Shelf: CRCA ▪ Legs: Aluzinc
Fuel	Firewood, bamboo, crop residue, other type of biomass
Life Span	10 years

The types of and levels of service provided by ICSs:

Level of Service:

Project Instances will provide ICS based on the Kuniokoa cookstove specifications as stated above, delivering a level of service at least equivalent to the baseline appliance (open fire used for household cooking). Each model of ICS distributed under Project Instance will be specified in the monitoring report.

Type of Service:

The Kuniokoa cookstoves will deliver a specified high-power thermal efficiency of at least 25% or greater and evidenced by a manufacturer's specification and WBT certificate. This is a significant improvement on the baseline open fire thermal efficiency default of 10% (as per AMS II.G).

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributor information

The information collected will be stored in which will serve as project database for project monitoring and sampling purposes.

1.12 Project Location

The GPA will distribute ICS to communities within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance is projected to be within the Highlands Region (Figure 2), and will include the following areas: SHP, WHP, EHP and JP (see Table 1 & Figure 2).

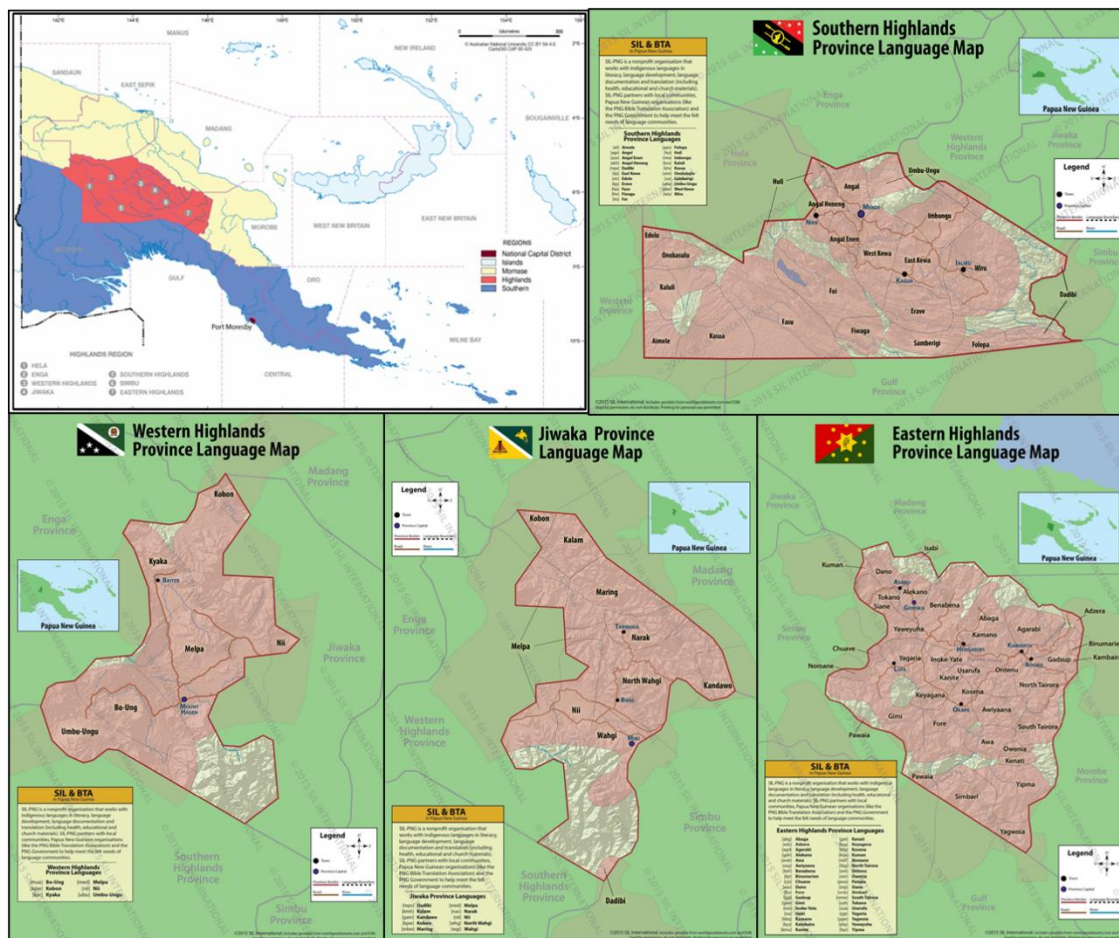


Figure 2. Map of Papua New Guinea including the Highlands Region (top left) and maps of the four different provinces included in the first project instance, Southern Highlands Province (top right), Western Highlands Province (bottom left), Jiwaka Province (bottom center) and Eastern Highlands Province (bottom right). Maps extracted from www.pnglanguages.sil.org

Table 1. The Geographic coordinates of the four provinces involved as the first project instances of the group activity in Papua New Guinea.

S/N	Location	Latitude	Longitude
1.	Southern Highlands Province	6.4179° S	143.5636° E
2.	Eastern Highlands Province	6.5862° S	145.6690° E

3.	Western Highlands Province	5.8310° S	144.1720° E
4.	Jiwaka Province	5.8564° S	144.5218° E

1.13 Conditions Prior to Project Initiation

The conditions prior to project initiation are the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

The pre project devices are inefficient three stone fired or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves. The baseline scenario is the same as the conditions existing prior to the project initiation i.e., continue the use of traditional cook stoves (Please refer to section 3.4 – Baseline Scenario).

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of ICS in PNG. The project is a voluntary effort by the project proponent.

There is no negative concern made on the ICS project from the above law.

Additionally, TEM maintains a close relationship with PNG's National Authorities and constantly updates them on the project activity. As an example, on the 31/01/2023, TEM had consulted with PNG's Climate Change and Development Authority (CCDA) and Forest Authority (PNGFA). During the consultations, the draft PDD and Local Stakeholder Consultation Plan were provided to the leadership of the government bodies for their comments. However, no comments have been received.

The acknowledgment letters from the abovementioned authorities can be made available to validators upon request.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit.

Supply Chain (Scope 3) Emissions

The project activity involves the distribution of ICSs to households within PNG. The boundary of the emission reductions is limited to the decrease in firewood consumption of each household, hence there are no indirect upstream or downstream GHG emissions to the organization's supply chain.

TEM will continuously made know to the community and all end-users of the project that these ICSs will generate VCU's issued for greenhouse gas emissions, however, this section does not apply to the project activity.

1.17 Sustainable Development Contributions

The contributions of proposed project activity towards sustainable development are explained with indicators viz. social, economic, environmental, and technological well-being as follows:

Social Benefits:

- Reduces drudgery of locals of rural areas by reducing time spent for firewood collection and the wood loads the family members needs to carry. Reduction in firewood requirement would help in spending more time in productive activities such as education, employment etc.
- Improves overall health (particularly diseases related to respiratory system) of all the members of the family by reducing smoke/indoor air pollution in the kitchen.

Environmental benefits:

- Reducing rate of forest degradation /deforestation in the project area. Conservation of forest will enhance forest ecosystems and reduce soil erosion and loss of biodiversity as a consequence of deforestation.
- Reduce indoor pollution – improved cookstoves emits less smoke and reduces health impact (morbidity and mortality from respiratory diseases and other health hazards), as well as the medical expenditure involved.

Economic benefits:

- Employment opportunities for local communities involved in monitoring, training of users, undertaking periodic maintenance and post lifetime replacement.

Technological benefits:

- Introduction of new technology (i.e., ICS) to rural communities.
- Knowledge transfer to trainers including technicians for pertaining training to users over system operation and maintenance.
- Demonstration of a successful project at household level will create replication potential in other states and countries.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage as per the CDM Methodology: AMS-II.G. v11.0 “Energy efficiency measures in thermal applications of non-renewable biomass”.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

The project will only bring positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17. No potential negative environmental or socio-economic impacts have been identified for the project.

Our baseline analysis showed that only 10% of the 220 households surveyed from the local communities involved in this project activity purchased firewood from local vendors. During all the face-to-face LSCs, members of these communities expressed their consensus that the project did not impose any risk. When people were asked if this project was going to disturb those who

are making their living by selling firewood, the response was that with ICS, there are many benefits they can foresee than making a living out of selling firewood.

2.2 Local Stakeholder Consultation

The local stakeholder consultations (LSCs) took place in four provinces involved as the first project instances (SHP, WHP, JP, EHP) as indicated in Table 2.

Table 2. The dates and venues in which the face-to-face local stakeholder consultations took place within the proposed four provinces involved as the first project instances.

S/N	Province	Date
1	Southern Highlands Province	13/03/2023
2	Western Highlands Province	15/03/2023
3	Jiwaka Province	17/03/2023
4	Eastern Highlands Province	20/03/2023

Prior to the LSCs, TEM utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using:

- Newspaper Advertisements
- Letters of Notification to Provincial Governments
- Radio Broadcasts
- Invitation Letters to Churches
- Social Media Posting
- Webinar
- Email to PNG Australia Alumni Association (PNGAAA) members

All advertising outlets were done in a staggered format with the aim of keeping stakeholders informed and aware of the LSCs that would be happening within their provinces. For exact dates of advertisements being published, please refer to Appendix I: LSCs Outreach.

The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include:

- Representatives from the provincial administration
- Representatives from Women Groups
- Representatives from Churches
- Ward councilors

- Heads of households both male and female
- Key informants such as teachers, community health workers etc.
- Improved Cookstove (ICS) users (SHP & WHP only)
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. Including those who commuted into Mendi from various districts, there were public servants such as nursing officers, police, teachers and church elders and pastors present during the consultation.



Figure 3. Pictured above, a demonstration by Ms. Magret Yangpin and Mr. Richard Manda – SHP Field Assistants

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by TEM did a power point presentation after welcoming and acknowledging all the stakeholders that were present. The PowerPoint slides were prepared in English, and explained in *tok pidgin*, the common language where most could understand well as those participated came from different local dialects.

Attendees were briefed of the purpose of the meeting, and informed of the:

- Carbon cycle and the important role it played within their forests.
- Findings of the Pilot Project conducted in SHP and WHP
- The Verra registration process
- The role of carbon finance in the project activity
- Indicative timeline and implementation plan of the project

Two field assistants were then invited to demonstrate and to explain how the improved cookstove is used using the sample that was exhibited during the LSC. That was further reinforced by one of the users who was also invited to give her experience on how she is currently using the ICS and explained the benefits her family is experiencing.

After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent answers provided by TEM, please refer to Appendix II: Stakeholder Feedback.

After the discussions, there were about **29** feedback forms that were received through the feedback forms. These feedback forms may be made available to validators upon request.

Western Highlands Province LSC

A total of 61 people attended this LSC on the 15th of March 2023 held at Kuri Lodge in Mt Hagen, WHP. Participants who attended came from different backgrounds and electorates or districts. The LSC was also covered by Radio Eagle (local radio station) that was present all throughout the presentation.



Figure 4. Group photograph at the Western Highlands Province LSC

Prior to the introductions and presentation, Pastor Enoc Wan of Hagen Central – Pentacoastal church opened the session with a word of prayer. Mr. Michael Kisombo then introduced the project and invited his field assistants to introduce themselves. He then proceeded with the presentation using *tok pidgin*.

He gave a brief introduction of the flow of his presentation and explained the purpose of the meeting. He then introduced the project activity before giving an overview of the carbon emissions and the forests' role in restoring carbon and its impact if most forests are destroyed. Following the background of the project, he explained the pilot project conducted in WHP and the results attained from the survey and invited end users (from the pilot project) present to explain how they are currently using the improved cookstoves to date and its impacts on them.

To note, the presentation was made to be more simplified so that more participants would easily understand its contents. The change was made upon reflection of feedback received during the SHP LSC where participants asked for technical concepts to be broken down to commonly used languages and analogies. For example, the concept of carbon emission destroying the ozone layer; Mr. Kisombo would explain it as 'bad air that destroys the blanket protecting sun and earth' and so on.

Two volunteers in the group gave their reflections and how they are using the cookstove. Both spoke highly of the impacts the ICS had on their families. Michael then asked Tony (one of the fields assistants) to demonstrate how the improved cookstove is used. After that presentation, Mr. Kisombo proceeded with the explanation of the Verra registration process and role of carbon finance and continued to explain the large-scale project, its indicative timeline and implementation schedule.

After the presentation, stakeholders were given the opportunity to ask questions and to give their reflections followed by a closing prayer by Pastor Enoc. 29 feedback forms were received through the feedback forms out of the 61 participants.

Jiwaka Province LSC

More than 90 people attended this LSC on the 17th of March 2023 held at Jiwaka Mission Resort (JMR) in Banz, Jiwaka. During initial consultations to invite the provincial administrator to the LSC, the provincial administrator showed great support for the project and was able to communicate widely to his people to participate. Due to work commitments, he sent his Deputy Administrator to attend the local stakeholder consultation instead.



Figure 5. Presenter Michael Kisombo (right) and the Deputy Administrator (Left)

Mr. Michael Kisombo led the discussions with a welcome and thanked everyone who came to Jiwaka Mission Resort for the local stakeholders meeting. He also did the presentation in *tok pidgin* – aided by the slides on power-point that was used in two other provinces. Similar to past LSCs, the topic on carbon cycle and the destruction of the ozone layer was simplified to the level of the participants. This topic was further elaborated on and supported by a few participants who had wider experience working or teaching in schools in this area of specialty.

The pilot in SHP and WHP and the coordinators of these projects, Mr. Patrick Wapiyu and Arnold Terry, were also on hand to explain the results of the pilot study and demonstrate how the improved cookstove was used. After these discussions, Mr. Kisombo continued with the presentations on the purpose of the stakeholder meeting, the verra registration process, carbon finance and the sustainable goals that this project is meeting. Towards the end, the indicative timeline and implementation schedule was explained before allowing for question-and-answer session.

There were about **29** feedback forms received through the feedback forms out of the 90 registered participants.

Eastern Highlands Province LSC

Around 29 people attended this LSC on the 20th of March 2023 held at Steakhouse Goroka in Goroka, EHP. The low turn-out rate may be attributed to the project being very new to the local stakeholders. Most stakeholders who attended learnt of the event word of mouth or through friends' network and face-book post. Despite a lesser number the discussions were fruitful. Almost all agreed that they see there is no harm and supported the project.



Figure 6. Community members involved in the EHP LSC at Steak House Goroka.

Similar to the first three LSCs, the presentation was conducted by Mr. Michael Kisombo in *tok pidgin* – as the common language where most could understand well. Michael followed the same presentation he did in the other three provinces. The sample ICS that was used for this LSC was displayed and a demonstration was done on how the ICS would be efficiently used. Although it was the first presentation made to the stakeholders in the Eastern Highlands Province, everyone who attended was positive with the project and supported the project activity.

After the discussions, there were about **24** feedback forms received from the 29 participants.

At the end of the LSCs, stakeholders were informed of the project's email (pngcookstoves@tem.com.au), as well as direct contact details to the local project manager and lead project manager in TEM. In all marketing documents, the project email and website were also published, to ensure stakeholders had multiple avenues to reach out to TEM with feedback, allowing for on-going communication.

For more information, a Local Stakeholder Consultation Report is available upon request of the validator.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

The project activity is currently undergoing the listing procedure and Public Comment on the Verra Platform. This section will be updated upon conclusion of the Public Comment Period.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

- VMR0006: Methodology for installation of High Efficiency Firewood Cookstoves v1.1.
- CDM Methodology: AMS-II.G. v13.0 “Energy efficiency measures in thermal applications of non-renewable biomass”.
- CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, v4.0

3.2 Applicability of Methodology

S/N	Applicability Criteria	Applicability
1	Project activities shall be implemented in domestic premises or in community-based kitchens;	Applicable The project activity will be implemented in domestic premises or in community-based kitchens.
2	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	Applicable The thermal efficiency of the stoves exceeds the required thermal efficiency of at least 25%.
3	Both 'Projects' and 'Large Projects' can use the methodology.	Applicable The project activities will fall within either 'Projects' or 'Large Projects', therefore this criterion is met.
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics; and	Applicable Non-renewable biomass has been used in the project region since 31 December 1989.
5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	Not Applicable Biomass residues are not processed as a fuel for this project.
6	The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices	Applicable

	including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	In order to avoid double counting of emission reductions through the method of distribution, each ICS will carry a unique identifier number and the distribution of each device is logged onto a database which includes unique information for each recipient (i.e. National identification number, geographic coordinates, etc.).
7	The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	Applicable To avoid double counting of emission reductions, rights to the emission reductions are transferred via a signed agreement stating that ER/VCUs are the sole property of TEM.

Evidence that non-renewable biomass has been used in the project area since 1989

Fuelwood is the primary energy source for cooking and heating in the highlands of Papua New Guinea, especially in the highlands where 40%¹ of the island's population lives and has become an important component of the domestic energy market; it is also a very easy market to enter as a small business. Although alternate energy sources have been made available within the region (electricity, kerosene, gas), these sources had not been made easily available or affordable. There was also an attempt to establish the common use of charcoal stoves into the community in the 1980s, however, the attempt was unsuccessful². As such, fuelwood has remained one of the major sources of domestic energy within PNG since.

Therefore, it can be concluded that non-renewable biomass has been used in the project region since December 1989.

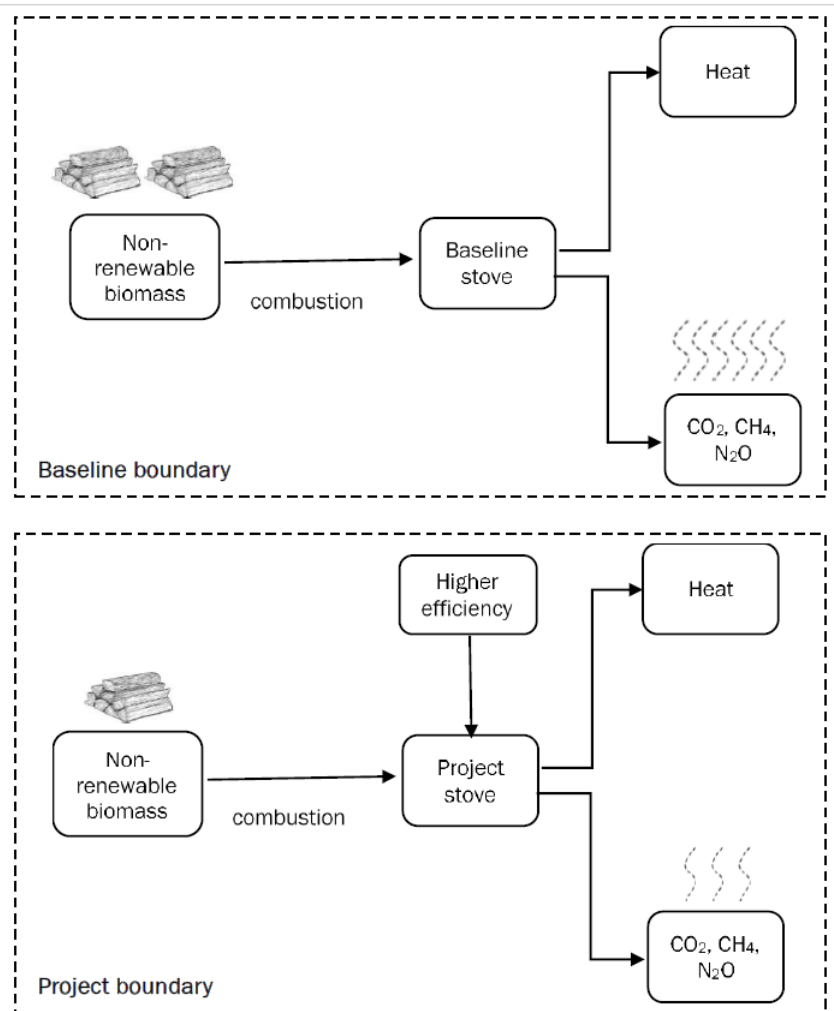
¹ Hanson, L., Allen, B., Bourke, M., & McCarthy, T. (2001). Papua New Guinea Rural Development Handbook. (1st ed.) Australian National University.

² Konedobu, P.N.G. : Energy Planning Unit, Department of Minerals and Energy, 1982

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of nonrenewable biomass/Fossil fuel	CO ₂	Yes	Major Source
		CH ₄	Yes	Major Source
		N ₂ O	Yes	Major Source
		Other	No	No other source considered
Project	Emission from use of nonrenewable biomass	CO ₂	Yes	Major Source
		CH ₄	Yes	Major Source
		N ₂ O	Yes	Major Source
		Other	No	No other source considered

The baseline and project boundaries are presented below:



Within the project boundary, less biomass burning gases and particulates are emitted.

3.4 Baseline Scenario

As per the VMR0006 methodology, the baseline scenario is the continued use of non-renewable wood fuel (firewood/charcoal) or fossil fuel (coal/kerosene) by the target population to meet similar thermal energy needs as provided by project cook stoves in absence of project activity.

Project Instances to be included into this grouped project activity shall meet the baseline scenario as mentioned above. For first project activity instance, the baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

3.5 Additionality

The methodology VMR0006 v1.1 uses activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in the host country of this project ensuring distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or regulatory framework, or for the UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by TEM.

Step 2: Positive List

The applicability conditions of this methodology represent the positive list. The project meets all the applicability conditions as described in Section 3.2. The project also meets the following conditions:

1. The project distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits.
2. Project activities are not implemented as part of government schemes nor is the project supported by multilateral funds.

The project can therefore be deemed as additional.

3.6 Methodology Deviations

The project did not apply any methodological deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

4.2 Project Emissions

The methodology does not account for baseline emissions separately, but instead quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-II.G. v11.1

4.4 Net GHG Emission Reductions and Removals

As per VMR0006, the net GHG emission reductions are calculated by applying Equations 1 and 2 for project activities replacing baseline stoves using non-renewable biomass (firewood/charcoal):

$ER_y = \sum_i \sum_j ER_{y,i,j}$		[Equation 1]
i	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace three stone fires.
j	=	Indices for the situation where there is more than one batch of improved cookstove of type i.
ER _{y,i,j}	=	Emission reductions by improved cookstove of type i and batch j during year y in tCO ₂ e.
l	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace three stone fires.

j	=	Indices for the situation where there is more than one batch of improved cookstove of type i.
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$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times fNRB_y \times (EF_{wf,CO2} + EF_{wf,nonCO2}) \times N_{y,i,j} \times 0.95$		[Equation 2]
$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j.
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default: 0.0156TJ/tonne).
$fNRB_y$	=	Fraction of woody biomass that can be established as non-renewable biomass.
$EF_{wf,CO2}$	=	CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel: 112 tCO2/TJ).
$EF_{wf,nonCO2}$	=	Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel: 26.23 tCO2/TJ).
$N_{y,i,j}$	=	Number of improved cookstoves of type i and batch j operating during year y.
0.95	=	Discount factor to account for leakage.

The quantify of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves can be estimated using Equation 3:

$B_{y,savings,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}} \right)$		[Equation 3]
B_{old}	=	Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of Bold can be sourced from historical data or baseline surveys. Alternatively, a default value of 0.5t/capita/year may be used.
η_{old}	=	Efficiency of baseline cookstove
$\eta_{new,i,j}$	=	Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT).

		Alternatively, efficiency may be determined using Equation 5.
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The efficiency of the project stoves can be estimated using Equation 5:

$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$		[Equation 5]
η_p	=	Efficiency of project cookstoves (fraction) at the start of project activity.
$(DF_n)^{y-1}$	=	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
0.94	=	Adjustment factor to account for uncertainty related to project cookstove efficiency test.

Where the project households continue to use baseline cookstoves along with improved cookstoves, B_{old} shall be adjusted ex-post based on the percentage of project households found to continue such practice according to Equation 6. For such cases, the quantity of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove.

$B_{old,adjusted} = B_{old} \times (1 - \mu_y)$		[Equation 6]
$B_{old, adjusted}$	=	Adjusted B_{old} to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device).
μ_y	=	Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves.

A net gross leakage factor of 0.95 is applied to $ER_{y,i,j}$.

ER Estimation for Project Instance:

For ex-ante calculation purpose, the assumption below is applied for first project activity instance:

- The first Project Instance installs 40,000 ICS in the first year, followed by another 40,000 ICS in the next, and finally 20,000 ICS in the third year to make up 100,000 ICS throughout the crediting period.
- The end date of installation of project activity instance is considered as operational date of that project activity instance.
- The life span of ICS is 10 years; thus, the operational lifetime of each project activity instance is taken as 10 years.
- Annual stove loss rate is estimated at 5%. This value will vary during actual verification based on survey results.
- $B_{old,adjusted}$ is fixed as 5.56 tonnes/year as per TEM's Baseline Study of the Highlands region.
- $f_{NRB,b,y}$ is fixed as 0.95 as per TEM's literature review and baseline calculations.

Determination of number of ICS operating during Year y:

$$N_{y,i,j} = 40,000 \times 0.95$$

In Year 2, another 40,000 ICS will be added, hence the calculation will reflect as:

$$N_{y,i,j} = (40,000 \times 0.95) + 40,000$$

In Year 3, another 20,000 ICS will be added to make up the 100,000 ICS, hence the calculation will reflect as:

$$N_{y,i,j} = (78,000 \times 0.95) + 20,000$$

Moving forward, the loss factor will be applied yearly to the total ICSs tabulated in Year 3.

Hence, the number of ICS operating during the crediting period is estimated below:

Year (y)	$N_{y,i,j}$
1	40,000
2	78,000
3	94,100
4	89,395
5	84,925
6	80,679
7	76,645

8	72,813
9	69,172
10	65,714

Determination of Efficiency of ICS during Year y:

$$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where:

$$\eta_p = 0.513^3$$

$$DF_n = 0.99$$

The efficiency of ICS during year y is stated below:

Year (y)	$\eta_{new,y,i,j}$
1	0.48
2	0.48
3	0.47
4	0.47
5	0.46
6	0.46
7	0.45
8	0.45
9	0.44
10	0.44

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y:

$$B_{y,savings,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}} \right)$$

As the project considers the possibility that there will be a continued use of baseline cookstoves along with improved cookstoves, the following equation is also applied:

³ As per manufacturer specifications, Kuniokoa cookstove with pot skirt

$$B_{old,adjusted} = B_{old} \times (1 - \mu_y)$$

Where:

$$B_{old} = 5.56 \text{ tonnes/year}$$

$$\mu_y = 0.1$$

$B_{old, adjusted}$ is then calculated as 5 tonnes/year.

Hence, the calculation for woody biomass over the crediting period is stated below:

Year (y)	$B_{old, adjusted}$	η_{old}	$\eta_{new,i,j}$	$[1-(\eta_{old}/\eta_{new,i,j})]$	$B_{y,savings,i,j}$
1	5.00	0.1	0.48	0.79	3.97
2	5.00	0.1	0.48	0.79	3.96
3	5.00	0.1	0.47	0.79	3.95
4	5.00	0.1	0.47	0.79	3.93
5	5.00	0.1	0.46	0.78	3.92
6	5.00	0.1	0.46	0.78	3.91
7	5.00	0.1	0.45	0.78	3.90
8	5.00	0.1	0.45	0.78	3.89
9	5.00	0.1	0.44	0.78	3.88
10	5.00	0.1	0.44	0.77	3.87

Determination of Emission Reductions:

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB_y} \times (EF_{wf,CO2} + EF_{wf,nonCO2}) \times N_{y,i,j} \times 0.95$$

Where:

$$NCV_{biomass} = 0.0156 \text{ TJ/tonne of wood}$$

$$f_{NRB,b,y} = 0.95$$

$$EF_{wf,CO2} = 112 \text{ tCO}_2\text{e/TJ}$$

$$EF_{wf,non-CO2} = 26.23 \text{ tCO}_2\text{e/TJ}$$

The calculation for emission reductions over the crediting period is shown below:

Year (y)	$B_{y,savings,i,j}$	$NCV_{biomass}$	$f_{NRB,b,y}$	$EF_{wf,CO_2} + EF_{wf,nonCO_2}$	$N_{y,i,j}$	$*0.95$	$ER_{y,i,j}$
1	3.97	0.0156	0.95	138.23	40,000	0.95	308,758.98
2	3.96	0.0156	0.95	138.23	78,000	0.95	600,488.88
3	3.95	0.0156	0.95	138.23	94,100	0.95	722,497.00
4	3.93	0.0156	0.95	138.23	89,395	0.95	684,511.55
5	3.92	0.0156	0.95	138.23	84,925	0.95	648,500.54
6	3.91	0.0156	0.95	138.23	80,679	0.95	614,362.23
7	3.90	0.0156	0.95	138.23	76,645	0.95	582,000.05
8	3.89	0.0156	0.95	138.23	72,813	0.95	551,322.41
9	3.88	0.0156	0.95	138.23	69,172	0.95	522,242.39
10	3.87	0.0156	0.95	138.23	65,714	0.95	494,677.55
Total:							5,729,361.59
Annual Average:							572,936.16

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
Year 1	308,758.98	0	0	308,758.98
Year 2	600,488.88	0	0	600,488.88
Year 3	722,497.00	0	0	722,497.00
Year 4	684,511.55	0	0	684,511.55
Year 5	648,500.54	0	0	648,500.54
Year 6	614,362.23	0	0	614,362.23
Year 7	582,000.05	0	0	582,000.05
Year 8	551,322.41	0	0	551,322.41
Year 9	522,242.39	0	0	522,242.39
Year 10	494,677.55	0	0	494,677.55

Total	5,729,361.59	0	0	5,729,361.59
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5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	B_{old,i}
Data unit	Tonnes/year
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices.
Source of data	TEM (2023). Cookstoves Highlands region report
Value applied	5.56
Justification of choice of data or description of measurement methods and procedures applied	Baseline was conducted as per the latest version of the CDM Standard for sampling and surveys for CDM project activities and programme of activities provided by the methodology.
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	B_{old,adjusted}
Data unit	Tonnes/year
Description	Adjusted B_{old} to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove
Source of data	Calculated as per Equation 6 of the methodology
Value applied	5.00
Justification of choice of data or description of measurement methods and procedures applied	Adjustment made to B_{old} to account for the possibility of open fire cooking practices being used in up to 10% of the end users during the project lifespan in addition to ICS.
Purpose of Data	Calculation of emission reductions

Comments	N/A
Data / Parameter	$f_{nrb,b,y}$
Data unit	Fraction
Description	Fraction of woody biomass that can be established as nonrenewable biomass.
Source of data	TEM (2023) f_{NRB} calculation based on Literature Review_Final
Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per the CDM Methodological Tool 30: Calculation of the fraction of non-renewable biomass. Audited by the Head of Forestry of PNG University of Technology.
Purpose of Data	Calculation of emission reductions. The parameter shall be determined ex-ante.
Comments	N/A

Data / Parameter	$NCV_{woodfuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass, renewable biomass, briquettes, or pellets used in project devices
Source of data	Methodology Default
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default for wood fuel, 0.0156 TJ/tonne, based on the gross weight of the wood that is 'air-dried' may be used if fuel used in project device is woody biomass/renewable biomass.
Purpose of Data	Calculation of emission reductions.
Comments	N/A

Data / Parameter	$EF_{WF,CO2}$
Data unit	tCO _{2e} /TJ

Description	CO2 emission factor of biomass that is substituted or reduced.
Source of data	Methodology Default
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of Data	Calculation of emission reductions.
Comments	N/A

Data / Parameter	$EF_{WF,non-CO2}$
Data unit	tCO _{2e} /TJ
Description	Non-CO2 emission factor of biomass that is substituted or reduced.
Source of data	Methodology Default
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of Data	Calculation of emission reductions.
Comments	N/A

Data / Parameter	$(DF_n)^{y-1}$
Data unit	Fraction
Description	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction).
Source of data	Methodology Default
Value applied	0.99

Justification of choice of data or description of measurement methods and procedures applied	Methodology Default. This parameter shall be determined ex-ante.
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	N/A

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Value applied	10
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of emission reductions and Determination of Crediting Period.
Comments	N/A

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Methodology Default
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of Data	Calculation of emission reductions.

Comments	N/A
Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of ICS at the start of project activity
Source of data	ICS Manufacturing Data and Specifications Sheet
Value applied	0.513
Justification of choice of data or description of measurement methods and procedures applied	As per WBT certificate provided by ICS manufacturer.
Purpose of Data	Calculation of emission reductions.
Comments	N/A

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type I and batch j operating during year y
Source of data	Monitoring survey, Purchase Invoices
Description of measurement methods and procedures to be applied	The total number of devices in operation compared to the total number distributed (according to the monitoring database) will be surveyed. Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	Yearly
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 40,000 ICS in Year 1
Monitoring equipment	Monitoring Survey

QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter.
Purpose of data	Calculation of emission reductions.
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	N/A

Data / Parameter	$\eta_{new,i,y}$
Data unit	Fraction
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity.
Source of data	Calculated as per Equation 5 of the methodology
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: "Efficiency of the improved cookstoves to be estimated using equation 5 of applied approved methodology where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored"
Frequency of monitoring/recording	Yearly
Value applied	Year 1: 0.48 Year 2: 0.48 Year 3: 0.47 Year 4: 0.47 Year 5: 0.46 Year 6: 0.46 Year 7: 0.45 Year 8: 0.45 Year 9: 0.44 Year 10: 0.44
Monitoring equipment	N/A

QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	As per equation 5 of VMR0006, version 1.1
Comments	N/A

Data / Parameter	Life Span
Data unit	Years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency.
Source of data	Manufacturer specification sheet
Description of measurement methods and procedures to be applied	The data source as per manufacturer specification.
Frequency of monitoring/recording	Once at the time of installation of batch of type of stove.
Value applied	10
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reduction.
Calculation method	N/A
Comments	N/A

Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	This parameter should be monitored using one of the following methods: 1) If the baseline cookstoves are decommissioned and no longer used, as determined by the monitoring survey its value is 0 and B_{old}, adjusted is equal to B_{old}. 2) If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting evidence to determine the frequency of usage of both the improved cookstoves and baseline cookstoves.
Frequency of monitoring/recording	Yearly
Value applied	0.1
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual / biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 09.0. In the case the desired precision is not met, lower bound values shall be used against repeating the survey to determine the operational fraction of stoves.
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	N/A

5.3 Monitoring Plan

Sampling Plan

The sampling plan is intended to enable the discovery of unbiased and reliable estimates of monitored parameter values used in the GHG emission reduction calculations. The sampling plan design is based on the CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 4.0, and will be designed to monitor the parameters listed in Section above, which are required for calculation of the actual GHG emission reduction achieved by the project activity using ex post sampling survey.

The share of operating stoves and the continued use of pre-project devices will be determined based on sampling procedures as outlined below. The Project Participant will be responsible for conducting the sampling surveys and maintaining a database with all operating stoves.

No monitoring for leakage through competitive uses of biomass is required, as the parameter wood usage is multiplied by 95% to account for leakage.

As per the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04, the sampling plan is the following:

1) Sampling Design

Due to the large number of ICS envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of a project instance-wide Sampling Plan (by grouping and sampling across project activities) that is designed in line with the requirements of the “Sampling and surveys for CDM project activities and programme of activities”, version 04.0.

1.1) Objective and Reliability Requirements

Sampling activities are defined at the Grouped Project-level and the sampling plan presented here will apply to assess parameter values in Project Instances to be included in the Grouped Project, either individually or in groups.

Monitored Parameters

Parameter	Description	Parameter Type/Frequency
$N_{y,i,j}$	Number of project devices of type i and batch j operating during year y	Proportion/Biennially
$\mu_{y,i,j}$	Adjustment to account for any continued use of pre-pre-project devices during the year y	Proportion/Biennially

1.2) Target Population

- The target population for the proportion of ICS still in operation ($N_{y,i,j}$ and $\mu_{y,i,j}$) of this project are all households in the project database which are using fuel wood in ICS distributed under the project for cooking.
- The target population for pre project appliances ($\mu_{y,i,j}$) is the set of old stoves still in use under the project database.

1.3) Sampling Frame

The projects are to be implemented in rural/semi-urban areas; thus, it is expected that the geographical locations do not have influence on the parameter of interest. Therefore, all above mentioned parameters can be assumed to be highly homogeneous for each ICS model regardless of how the end user group and distribution/installation location is defined.

1) Sampling frame for proportion of ICS in operation ($N_{y,i,j}$)

The sample frame refers to all the information sources on the Database. There are two primary mechanisms for data collection: the Registration Process for newly distributed/installed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the ICS. The detailed information collected from Registration Process is used to populate the stoves Database and the Monitoring Survey follows “Sampling and Surveys for CDM Project Activities and Programme of Activities”, version 04.

As explained below (on section “Sampling Method”), to take the different characteristics of different project instances Implemented and ICS models into consideration, Project instances shall be grouped together to create a Primary Sampling Unit, which is homogenous. As per EB 86 Annex 04, Appendix-2, paragraph 1, for the use of a single sampling plan covering a group of projects, provided the homogeneity of population can be demonstrated, or differences are considered in the sample size calculation, a 95/5 confidence/precision is applied for biennial sampling.

The first step is to identify the Primary Sampling Units. Primary sampling units are project instances, which have:

Identification of Primary Sampling Units:

Primary sampling units are project instances, which have:

- The same ICS models

That is project with the same ICS model can therefore be grouped together and form a Primary Sampling Unit.

- Adjustment to account for any continued use of pre-project devices during the year($\mu_{y,i,j}$)

In line with applied approved methodology AMS-II. G, as installing data logger is not practical and if any use of pre project device can be monitored in a common survey with other monitoring parameters; therefore, a random sub-sample within the common survey can be taken to determine continued use of old cookstoves and its proportional usage by including suitable questionnaire.

The outcome of the survey could lead two results:

- 1) The project ICS was completely discarded.
- 2) The old stoves are used along with project ICS.

In the first case, it will be simple multiplication of fraction of total number of project ICS displaced by old cookstoves by total number of cookstoves in project, to achieve precise results based on survey result sample size calculation can be repeated.

However, in the second case, surveys may be conducted if the use of data loggers to record the continued operation of baseline devices is demonstrated to be not practical, for example when the baseline device is the three-stone fire. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidence to determine the frequency of usage of both the project devices and baseline devices.

For example, if there was only one baseline cookstove device per household and its use during the project lifespan continues along with the project stove to meet 25% of the cooking needs of the household in which case the $\mu_{y,i,j}$ parameter will be 0.25. Where a more precise data is available i.e., the thermal capacity of the project and pre-project devices and respective utilization hours, a weighted average adjustment factor may be used.

1.4) Sampling Method

For each survey, the sampling method for monitored parameters ($N_{y,i,j}$ and $\mu_{y,i,j}$) is determined by using a Simple Random Sample of the total population of distributed ICS from records generated in the monitoring database and based on the precision requirements listed above, and as required by the methodology and guidance in the Sampling Standard.

To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the Project Participant.

To determine the parameters, sampling will involve the following approaches:

$N_{y,i,j}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use.

$\mu_{y,i,j}$: Pre project device only is in use then fraction to be used to calculate total number, however if pre project device is used along with project ICS, proportion of usage of each will be determined by cooking habits evaluated by survey questionnaire during the monitoring period.

Using the formulas in the section “Sample Size” below, the Project Participant will randomly sample the required number of ICS from the primary sampling units. It is important to note that for $\mu_{y,i,j}$ where partial usage of both old stoves and project ICS are observed, for each household under sample cooking habits must be taken into consideration.

1.5) Sample Size

The sample size for each monitoring activity will be the total number of ICS distributed and included in the monitoring database. The minimum sample size for each sample frame must achieve the 95/10 confidence/precision. For annual and 95/5 confidence/precision for biennial sampling. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error.

In line with sampling guidelines, all can be sampled in a single survey with a random sample of households using the above-described confidence/precision levels depending on annual or biennial monitoring frequency. The $N_{y,i,j}$ and $\mu_{y,i,j}$ requires proportion/percentage parameters.

As per Guidelines for Sampling and surveys for CDM project activities and programmes of activities, version 07.0 Appendix 1 paragraph 5, there are different ways available to obtain the estimates of the parameter of interest:

1. Refer to the result of previous studies and use these results.
2. In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates.
3. Use best guesses based on the researcher’s own experiences.

For the registration purpose of project, option 3 shall be applied. For the first monitoring period, values from a pilot shall be applied. For the following monitoring periods, the estimates shall be adjusted considering the results of the previous monitoring period(s) or the result from recent pilot study, which is conducted after the previous monitoring periods.

To estimate the sample size for parameters $N_{y,i,j}$ and $\mu_{y,i,j}$ the following equation is used:

$n \geq \frac{1.96^2 N \times p (1-p)}{(N -1) \times 0.1^2 \times p^2 + 1.96^2 \times p(1-p)}$		
n	=	Sample size
N	=	Population size (Total number of households/ICS)

p	=	Expected proportion
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In case the resulting sample size to achieve the desired confidence/precision levels is smaller than 30 ICS, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion in line with para 14 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 9.0.

TEM envisages distribution of 40,000 ICSs in Year 1. Hence, population size, N, is taken as 40,000 household/ICS (assuming 1 ICS per household).

With an expected ICS loss rate of 5%, it is expected that 95% of ICS will be operational, hence the expected proportion p for $N_{y,i,j}$ is taken as 0.95.

Oversampling will be conducted to ensure that the required level of precision is met, not only to compensate for any attrition, outliers or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required.

1.6) Data Recording

The distribution record including user information will be collected and recorded at the point of distribution by the staff or distributors. Alongside the unique serial number, information recorded will include:

- Name of cookstove user
- Address/GPS of cookstove household
- Phone number of cookstove user
- Cookstove model
- Date of distribution and installation
- Name of the staff that delivered and installed the cookstove.
- Photographic evidence

The information collected from the end-user will be transferred to an electronic database, which will be updated regularly. The distribution record carries all distribution information including the traditional stove type (charcoal stove/firewood stove) used prior to ICS installation, name of representative member of household, phone number (if available and user-permitted for sharing), address etc. Likewise, monitoring records are transferred from each monitoring organization, if any, to the PP. PP will ensure that appropriate records are maintained for the project activity.

A record keeping system and unique identification of the project is made possible by user's information collected through the End-User Agreement and compiled in distribution records. This will include end-user name, identification number and address/location of the user's house,

stove unique serial code and distribution date. The record keeping system will ensure that each ICS can be traced to the project activity to avoid double counting.

The Monitoring Plan applied in this project involves several key elements that ensure that the PP have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating emissions reductions on the basis of the amount of non-renewable biomass saved by the ICS in the project activity.

The key elements are the following:

- Project database management
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The TEM Country Manager will have overall responsibility for the implementation of the monitoring plan. The information collected will be stored in the electronic database and will be updated on ongoing basis by trained staff.

Data collection will be conducted by a dedicated monitoring team of trained individuals, from TEM. All technical staff responsible for installation and maintenance of the stoves will be trained in terms of the understanding the requirements on the monitoring system.

1.7) Organizational Structure

The project will be managed by TEM as the project participant. The roles and responsibilities are shown in the table below:

Person/Entity	Responsibility
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TEM Country Manager, Field Coordinator and Field Assistants	• Oversee operation of distribution centres; execution of set up activities; works with project manager on all planning. • Execution of set up activities such as recruitment and training, reporting of monitoring data. • Logistics Manager: planning; identification of target households; contractor management; overall day to day management of installation staff; weekly and monthly reporting. • Data administrators: monitoring database management; accounting; data reconciliations; monthly reporting • Pre- & Post-distribution data collection: conveying project messages; distributing the project; signing up householders wanting a stove; sign up data capture; distribution data capture • Distribution team: management of distribution process; ensuring quality stove distributions. • Monitoring team: gathering compliance monitoring data; gathering marketing data; data input
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TEM Head of Science and Sustainability	• Project planning and management; issue and risk management. • Oversight of management system & signoff on ICS procurement and monitoring reports, review of competencies of team members • Technical review team: technical review of process and documentation and monitoring reports. • Compliance Manager/Consultant: writing PD & monitoring reports, ensuring compliance with VCS rules.
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Elaboration of Tasks:

Person/Entity	Task	Responsibility
TEM Country Manager, Field Coordinator and Field Assistants	Manufacturing and logistics	Depending on the stove model, complete stoves or components for the stoves are: • Manufactured (some imported, others produced locally) by a stove manufacturer. • Stoves are distributed to warehouses within project region. • Coordinate the distribution of stoves to recipient households
TEM Country Manager, Field Coordinator and Field Assistants	Household identification	• Work with local community leaders and other partners to help identify recipient households suitable for the distribution of a stove.

		- Establish partnership with community leaders, NGOs and other local organizations, local partner will initiate a communication process to ensure that households understand the benefits of the stoves, that cultural issues are addressed and that users are trained in the optimal use and performance of the stove. - TEM PNG team prior distribution will visit recipient communities in each region and ensure recipients understand and acknowledge the conditions for participation in the project. Ensure each stove is assigned a unique distribution number chronologically to avoid any double counting through the use of an electronic database
TEM Country Manager, Field Coordinator and Field Assistants	Distribution	- TEM PNG team train stove distribution teams to distribute stoves. - Coordinate the receipt of stoves and components in the distribution process. - TEM PNG team will be trained in the distribution of the stove to a standardized delivery and communication procedure. TEM PNG team will be responsible for physically distributing the ICSs to the recipient household
TEM Country Manager, Field Coordinator and Field Assistants	Data Capture	- The information collected from the TEM PNG team during the distribution of ICS, from end-users and interactions on the monitoring phase of the project will be transferred to an electronic database, which will be updated regularly. The database will include a unique reference number for each stove to avoid double counting. - TEM PNG team checks the quality of ICS and record keeping work.

		If the work is satisfactory, distribution data is collected by the TEM PNG team
TEM Head of Science and Sustainability	Monitoring	Monitoring activities will be conducted as follows: • Surveys completed in the field by trained local monitoring teams. • Data captured by the monitoring teams is passed to the TEM Head of Science and Sustainability • Data is checked for completeness, consistency, and accuracy. • Project manager summarizes data in a report to TEM Head of Science and Sustainability • Writing of monitoring reports for each monitoring period. • Technical review by in-house technical team

1.8) QA/QC

In order to minimize sampling errors, such as non-responses and errors, the field team will practice over-sampling from the population to ensure a total number of respondents that meets the required level of precision. This will ensure the integrity of the sample group and maintains the randomness of participant selection. All samples' groups will be re-selected for each monitoring period / year, as appropriate for the parameter in question.

If the required level of precision is not achieved, then further surveys may be completed, or the lower bound value adopted in the calculations.

Where a survey may not be completed, or where there is a non-response, the reasons shall be clearly documented in the survey questionnaire.

APPENDIX I: LSC OUTREACH

1. Newspaper Advertisements:

There were two newspapers published nationally in PNG. As the project activity would be a grouped project activity, TEM saw the importance of announcing the to the public and informing stakeholders of the project, as well as brief them on the upcoming local stakeholder consultations happening in the Highlands Provinces. The following table shows the date of when these advertisements were published in the newspaper:

S/N	Publishing House	Advertisement Published on: (date)	Supporting Documentation
1	Post Courier Newspaper	14/04/2023	Invoices from respective newspapers for publishing, picture of advertisement in newspapers

2. Letters of Notification to Provincial Governments:

Letters of Notification were also hand delivered to provincial government administrative assistants to brief them on the project activity as well as the consequent co-benefits to their local community that came with the project. Together with the Letter of Notification, a project brief was also attached to aid their understanding. The following table shows the dates of when the letters of no-objection were received from the provincial governments:

S/N	Province	Document Received on: (date)	Supporting Documentation
1	Western Highlands Province	13/02/2023	Letter of No-Objection
2	Jiwaka Province	15/02/2023	
3	Eastern Highlands Province	16/02/2023	
4	Southern Highlands Province	08/03/2023	

During TEM's consultations with the provincial governments, the administrative assistants expressed a high level of interest in the project and were supportive of its implementation.

3. Radio Broadcasts

To further TEM's outreach into the Highlands Provinces regarding the LSCs, radio broadcasts were utilized. All radio broadcasts were done by the National Broadcasting Commission which then broadcasted TEM's announcements to individual provinces at a frequency of 1 time a day, over 10 days. Each province-level broadcast would then inform the specific province of the date, location and time of

which the LSC was happening within their area. The following table shows the start and end dates of these broadcasts:

S/N	Province	Start of Broadcast: (date)	End of Broadcast: (date)
1	Western Highlands Province	29/02/2023	12/03/2023
2	Jiwaka Province	29/02/2023	12/03/2023

The radio broadcast could only be done for 2 highlands provinces as it was found that the radio stations for the remaining two provinces had been shut down.

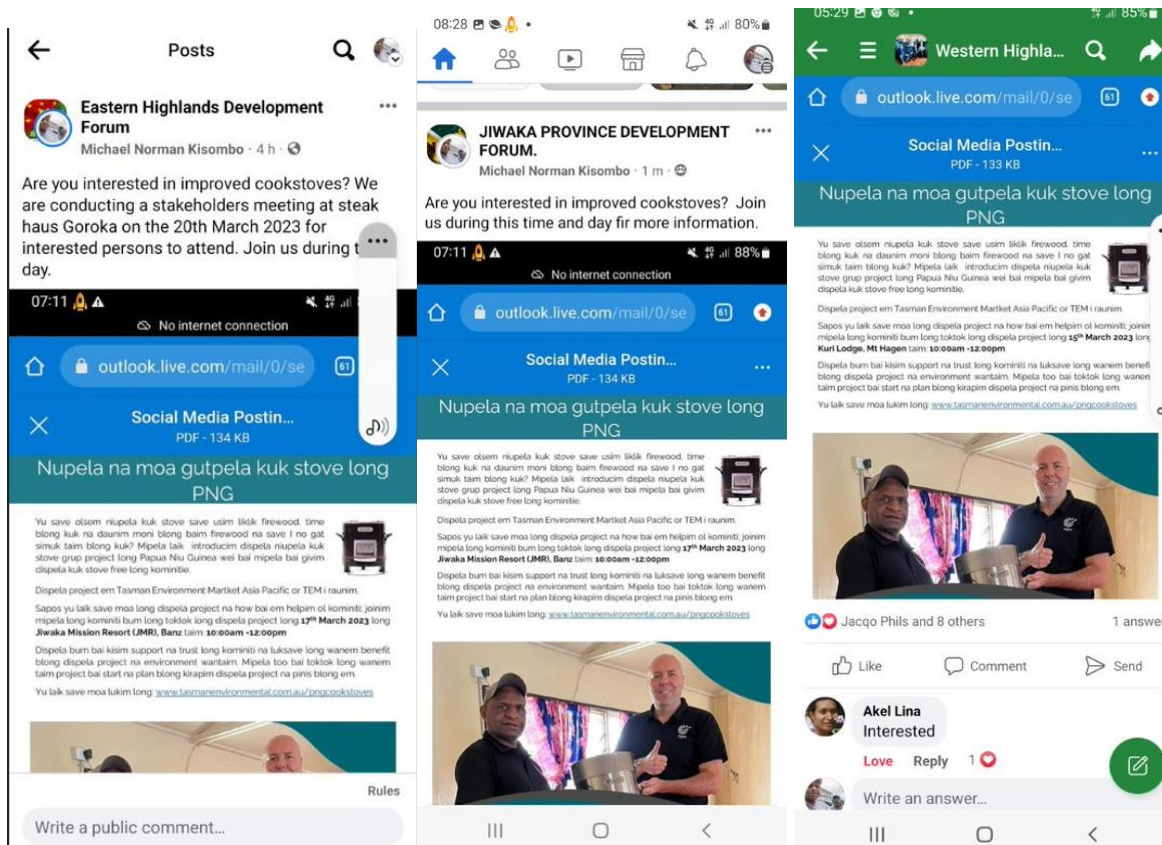
4. Invitation to Churches

TEM is aware that churches and church leaders play an important role in the local community within PNG. As such, invitations to churches were also hand delivered to churches within each province to invite church leaders and elders to participate in the LSC, as well as spread the word about the importance of the LSC within their community by word-of-mouth. The following table shows the dates these invitations were delivered to churches within the provinces:

S/N	Province	Delivery of Invitation: (date)	Supporting Documentation
1	Southern Highlands Province	10/02/2023	Acknowledgement of Receipt by Church Bodies: SHP
2	Western Highlands Province	15/03/2023	Acknowledgement of Receipt by Church Bodies: WHP, Jiwaka, EHP
3	Jiwaka Province	17/03/2023	
4	Eastern Highlands Province	20/03/2023	

5. Social Media Postings

TEM also understood that the majority of the stakeholders received news of events happening within their area through social media. For this, flyers depicting the details of the LSC, and introduction of the project were developed and posted on each province's dedicated Facebook development forums to further boost stakeholder awareness of the LSC. Postings of these flyers started on 13/03/2023.

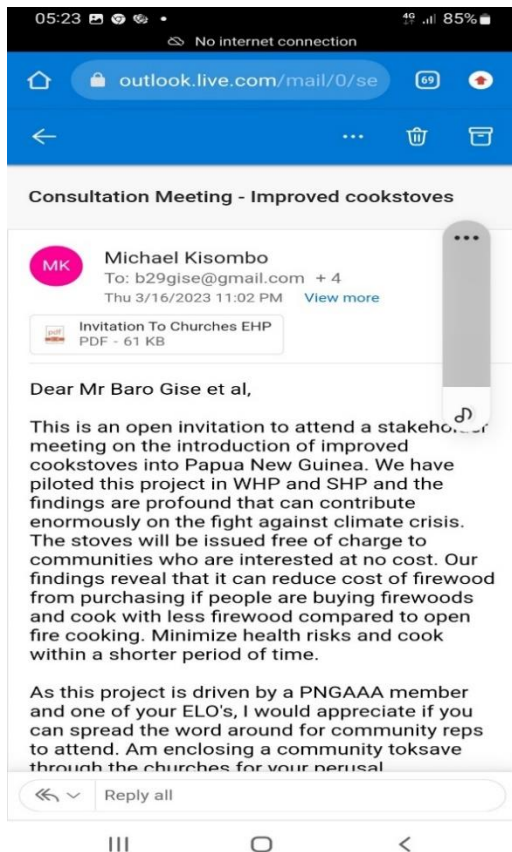
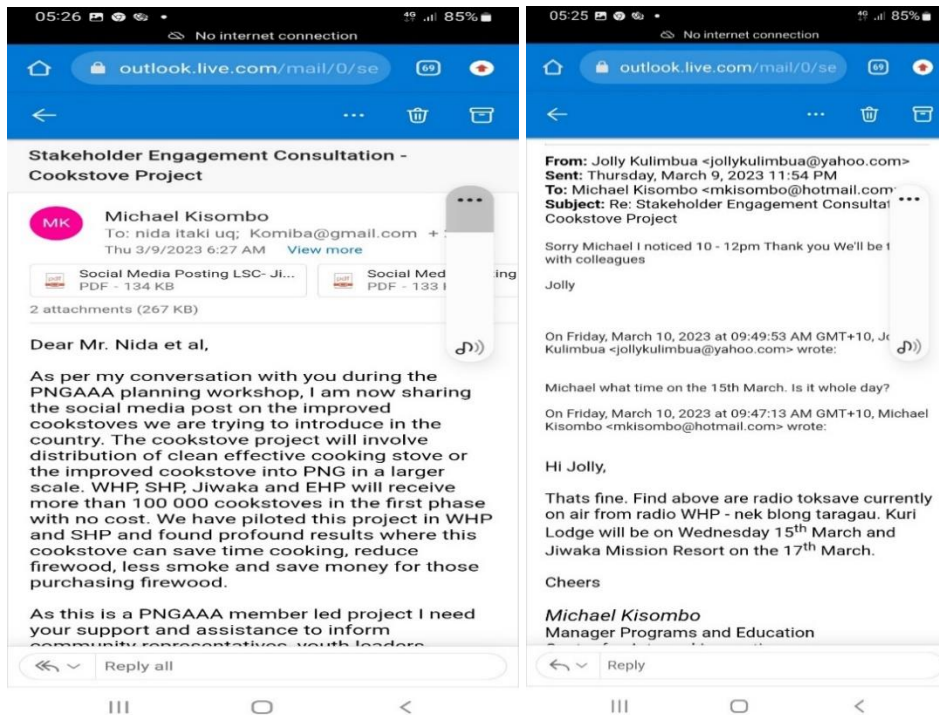


6. Email to PNG Australia Alumni Association (PNGAAA) members

From within TEM's local partner's connections, TEM also reached out to members from WHP and EHP via email. The table below shows the dates these emails were sent out:

S/N	Province	Date of Email Sent: (date)
1	WHP	09/03/2023
2	EHP	16/03/2023

Screenshots of the emails sent out can also be found below:



APPENDIX II: STAKEHOLDER FEEDBACK

1. Southern Highlands Province:

Questions:

S/N	Question	Answer
1	Ms Julie Ponaria (a teacher from SHP) enquired that there are many big countries contributing to the destruction of the ozone layer and now trying to give (us) cookstoves to stop carbon emissions. She asked if there is a story behind this small cookstove?	Yes, there are many industrial countries that are destroying the ozone layer because they have big factories and industries. We (people of PNG) also do so when cutting down trees for farming and cooking etc... This project is to reduce carbon emissions by using less firewood for daily needs. Hence, saving trees that consume carbon and restore them. That is the story behind this project.
2	Ps Joseph Yapasi of Tente Ward 2 commented that 50 000 cookstoves is not sufficient to cater for all households in Southern Highlands and there may be risks when distribution. Therefore, he suggested that distribution be more targeted. E.g distribute through church networks or distribute to a particular district first and later target other districts.	It was noted for TEM to decide.
3	Ms Regina Kamerepa pointed out that since the cookstoves are going to be issued free, people are likely to rush to get one. Hence, can also steal or pull them from the distributors during the distribution. And asked how TEM would manage that scenario?	This is a threat we are aware of and are working on a strategy on how this can be mitigated during distribution. Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.

2	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
3	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
4	Mr. Peter Wari who is in charge of climate change in the province acknowledged the project as it was really going to benefit the community. He further stressed about the importance of the project activity and encouraged the stakeholders to support the project.
5	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
6	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
7	Another youth mentioned that he provides firewood for three separate houses as most are windows who have no husbands. With the new cookstove, he believes will relieve him from time and struggles he is facing foraging for firewood for all he's mothers.
8	A Health worker emphasized that inhaling of smoke using normal open fire cooking is contributing to a lot of respiratory illnesses and with the introduction of the improved cookstove she believes will minimize such especially mothers and girls.
9	Besides the above reflections and questions, there were quiet a good number of people giving positive comments and supported the project activity. They even asked to increase the volume from 50 000 cookstoves to enough to cater for all households the province.

2. Western Highlands Province:

Questions:

S/N	Question	Answer
1	Mr. Frank Kuri Cr from Mul Baiyer asked, since it's going to affect the rural communities, can the councilors get involved?	Yes, Councilors will get involved to identify heads of households, but we do not want this project to be politicized, either be councilors or local MPs.
2	Mr. Nida L Taki, a lecturer at the Agriculture College, Mt Hagen asked if there are further studies made to determine the level of smoke that is given out during normal open fire cooking to see how much carbon emissions are produced.	That's an interesting question. However, this project used observation method to observe the level of smoke when cooking both

		during the control cooking test and open fire cooking.
3	Mr Dickson Wia stressed that firewood is not only used for cooking in the highlands but for other purposes too. Will this cookstove provide for all?	Yes, we use firewood for many other purposes like for heat, light, storytelling etc. This cookstove is for cooking and it can also heat up the house.
4	Rex Parak enquired if the design of the cookstove can be made locally?	The brand and model of cookstove was chosen specifically for its high efficiency. TEM will consider a local brand in the future if it fits the specifications required to yield the highest efficiency in the combustion of firewood.

Reflections:

S/N	Reflections
1	The District Administrator for Dei District thanked the team for introducing this new innovative idea of cooking as people in his electorate are running out of firewood. He welcomed the project initiative and asked to include all rural communities in his electorate.
2	Those who came from the urban centers also requested that they should be given priority as they are using a lot of money to purchase firewood and its already becoming a burden.
3	One of the participants mentioned that the project is very beneficial as it saves time foraging for firewood and a financial relief in buying firewood.
4	A youth mentioned that this project will benefit the youth as he sees that firewood are already running out.
5	Another supported by saying, the initiative taken by TEM is indeed very vital to reduce carbon monoxide as well as relieving economic burden and reduce health related issues. She suggested that it would be good to involve the councilors.
6	One said, this project must not stop after 10 years. It must continue till the world ends.

7	A health worker supported the project by saying, as a health worker this project if rolled out effectively will help reduce the number of sicknesses like pneumonia, chronic obstructive lung disease and eye diseases as well.
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3. Jiwaka Province:

Questions:

S/N	Question	Answer
1	Mr. Lumun Mingal asked how can PNG be compensated well since we are less carbon emitters yet are asked to further reduce the emissions through such cookstove project?	This cookstove project aims to contribute to the wellbeing of the stakeholders in PNG by lessening the burden of purchasing or collecting firewood and improving the respiratory health of end-users exposed to hazardous smoke from the burning of firewood. Cookstoves being distributed for free is paid for through carbon finance and is one method of compensating stakeholders using the carbon credit scheme.

Reflections:

S/N	Reflections
1	A local landowner thanked TEM for this initiative and welcomed everyone to JMR. Having said that he suggested that this project be based there as it is the model village for Jiwaka and must be made the main distribution point.
2	Another said, there are many NGOs like IOM, UNICEF, etc. who come and deliver similar community-based projects but we the people seem to sell those items. He was worried the cookstoves might be sold in a similar manner and appealed to other participants to discourage such practices.
3	A female teacher mentioned that as a teacher she moves from one area to another with kerosine stoves and with the introduction of the improved cooking stove, hers and other teachers would benefit greatly. Not only that but emphasized that they usually ask their students to bring firewood for them once a week and this project will ease that burden.

4	Another person suggested that every household in Jiwaka must receive a cookstove as they are facing difficulties in looking for firewood.
5	Sammy Buka of Banz mentioned that the project is a bottom up where local communities will benefit. He said it's a sickness free project.
6	There were other supportive comments like, TEM must continue this project to other provinces in PNG, delivered through churches, schools and LLG wards.
7	When asked if there are risks involved like those of you selling firewood, would this project affect you? Everyone said no, not at all as they see there is a lot of benefit to the rural communities.
8.	Someone also wanted more than 50 000 cookstoves in Jiwaka as they believe they are good people.

4. Eastern Highlands Province:

Questions:

S/N	Question	Answer
1	Mr Rex Yagusa mentioned that there is a high possibility of the cookstoves being stolen by thieves or destroyed by natural and man-made disasters and asked in such a case, how can we register these complains or cases?	Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	One of the ladies (video graphed) mentioned that this cookstove will lessen the burden for those mothers in the rural setting. She said if they don't sell, there is no money to purchase firewood so they would use plastics and cardboard to cook. This project will really relieve those struggling mothers living in the urban areas.
2	Another suggested having the distribution point centralized and allowing only genuine households to collect. Later field officers can be sent to their locations for registration such as GPS points etc.

3	Someone in the group mentioned that economic benefits are huge as they will be working as the project team to distribute.
4	Those who attended expressed support for the project and demanded if more is distributed in EHP given their population density.
5	No risks or harm mentioned when implementing this project.
6	One participant mentioned that tree-cutting is happening almost every day. More supplies to meet population will help reduce carbon emission.